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ICRA™

Privacy-Preserving Robotic Perception for Object Detection in Curious Cloud Robotics



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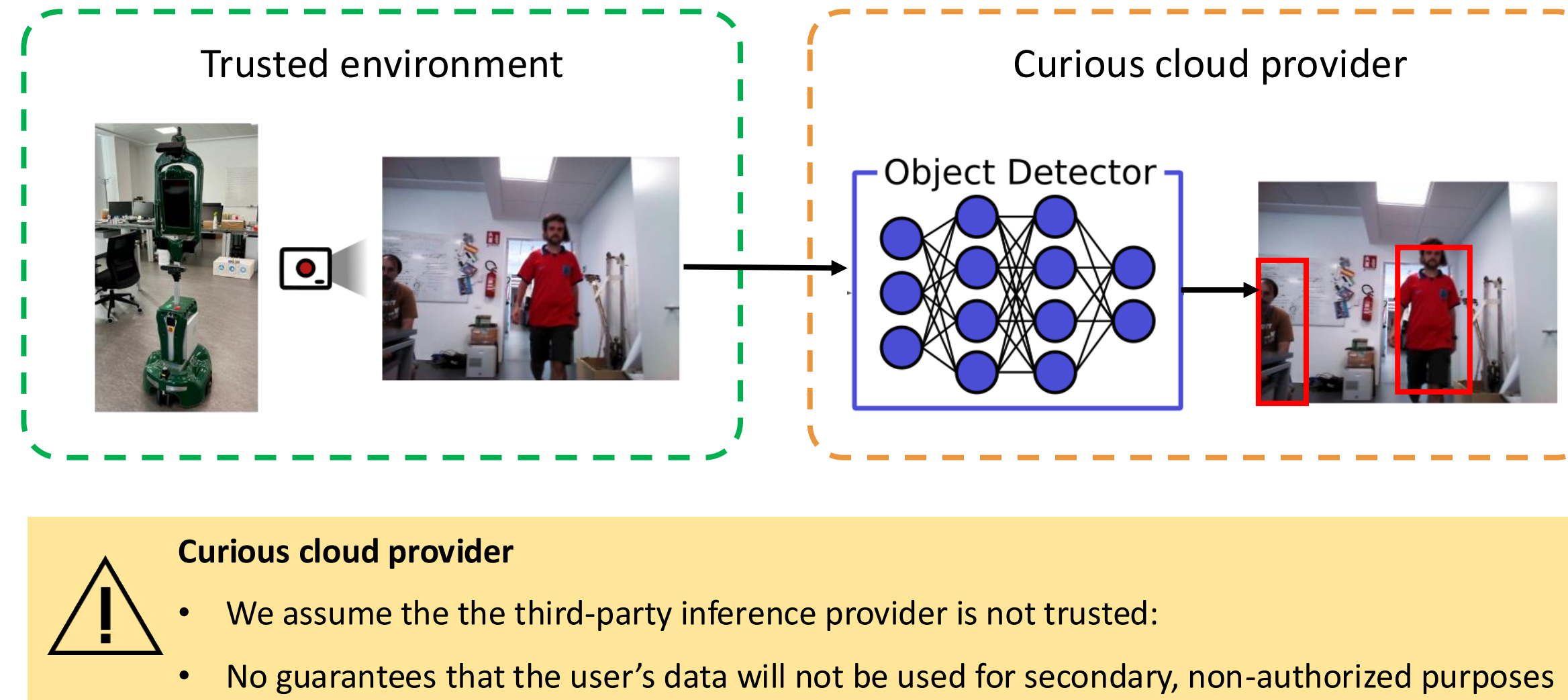
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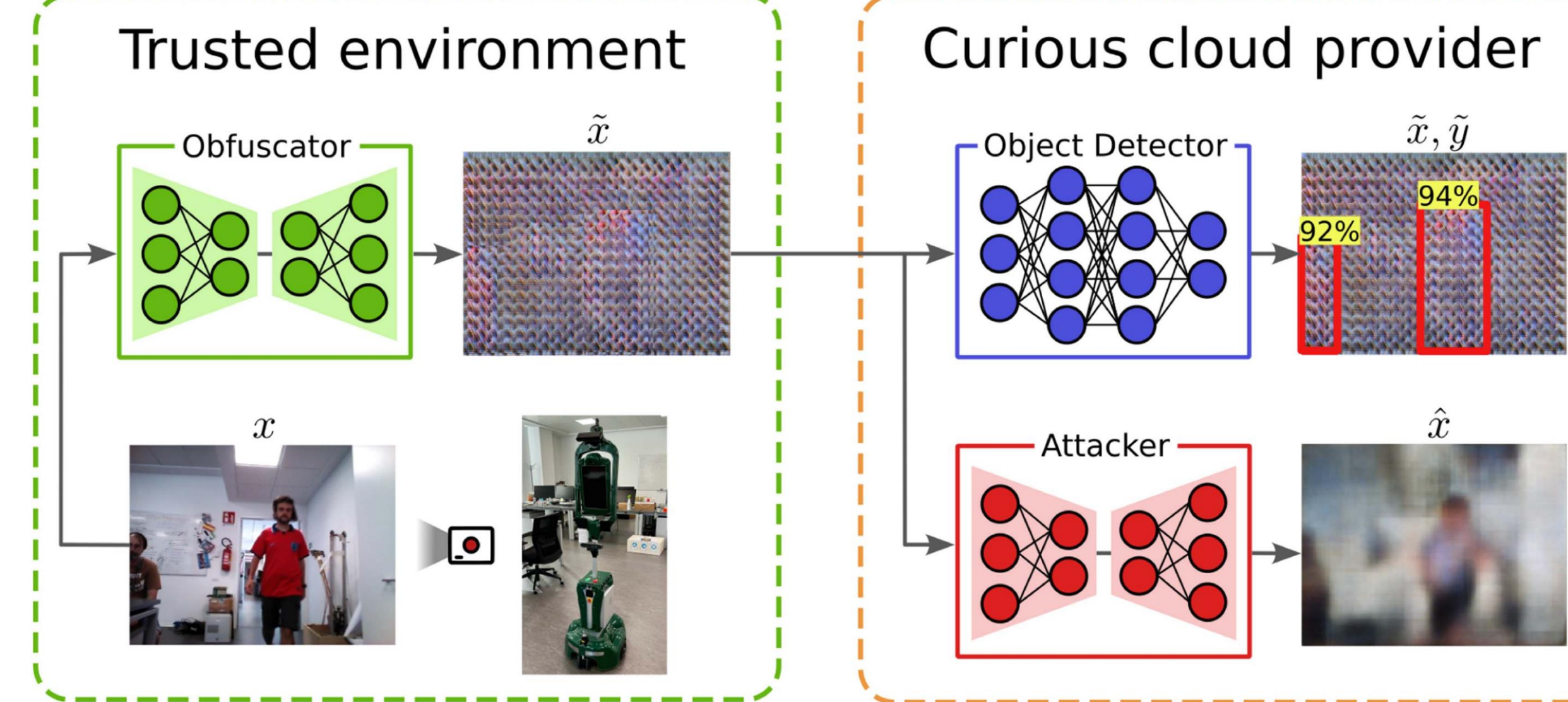
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Motivation

- **Low-power service robots** increasingly delocalize computationally heavy deep neural network (DNN) object detection inference to third-party providers
- Service robots operate in human-centric spaces like homes or hospitals, collecting high-resolution images that are **privacy-sensitive**^[1]
- Even if images are sent over an encrypted channel, cloud DNN inference needs clear data to operate
- **Can the we trust the provider with our data?**



Proposed architecture



- Equip the robot with a local and efficient **Encoder-Decoder Obfuscator**
- Transform images into noise-like data such that:
 - ✓ task-specific features are preserved: the **cloud-based TaskNet** can execute accurate object detection *without being retrained*
 - ✓ prevent adversarial reconstruction: an **attacker** struggles to reconstruct the original perception from the obfuscated one

Formalization

- **Obfuscation:** mapping image x to a low-dimensional latent space $z = e(x; \theta_e)$ and decoding it to an obfuscated output $\tilde{x} = d(z; \theta_d)$
- **Co-training optimization:** multi-objective loss balancing task accuracy and privacy

$$\mathbb{E}_{x \sim \mathcal{D}} \lambda \mathcal{L}_{task}(Y, \tilde{Y}) + (1 - \lambda) \mathcal{L}_{priv}^{\gg}(x, \tilde{x})$$

- **Attacker:** model Inversion Attack (MIA) network that tries to reconstruct x from $\tilde{x}$, trained with a reconstruction loss on images obfuscated with our method

$$\mathbb{E}_{x \sim \mathcal{D}} \mathcal{L}_{priv}^{\ll}(x, \hat{x})$$

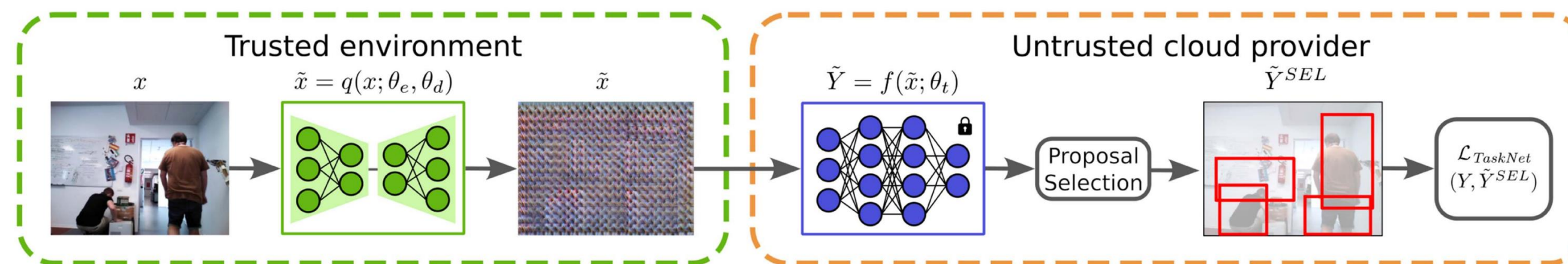
- **TaskNet:** fixed weights trained on a public dataset for object detection

Theoretical analysis

- Prior classification-focused methods^[2] achieved privacy by shrinking the bottleneck Z , forcing the network to drop task related features
- **Theorem 1 (Linear Detection-Aware Compression)**
In proposal-based object detectors, compressing Z below the rank S of the detector's backbone guarantees a non-zero task loss:
⇒ **full task performance recovery strictly requires $Z \geq S$**
- **Theorem 2 (Compression vs. Reconstruction)**
The attacker reconstruction loss decreases as Z grows, shrinking Z provides privacy but destroys detection capabilities:
⇒ **strategies solely relying on bottleneck tuning face an unresolvable trade-off**
- **Theorem 3 (Compression for detection and privacy)**
Zero task loss with a compressed bottleneck ($Z < S$) is possible if and only if the rank of the multi-head stacked backbone matrix is strictly less than S
⇒ **restricting gradients to a "weak set" of proposals leads to low-rank feature space**

Proposals selection

- Backpropagation is restricted to a sparse, **dynamically selected subset of proposals**: the obfuscator distills only the minimal features relevant to the task, discarding those that an adversary could exploit for reconstruction



Selection Algorithm: for each ground truth bounding box g select

- **Positive Proposals (P^{SEL}):** top p proposals that achieve the highest Intersection over Union (IoU) with g . These provide gradients to lock onto target localization and class correctness
- **Negative Proposals (N^{SEL}):** filter out positive proposals, enforce an IoU upper bound $\bar{\rho}$, and pick the top n proposals with the highest TaskNet confidence score. These force the obfuscator to actively suppress false background detections

Algorithm 1: Cotraining with Proposal Selection.

Input:

- $f(\cdot; \theta_f)$: a fixed and pretrained object detector
- $q(\cdot; \theta_q^e, \theta_q^d)$: an encoder-decoder with random weights
- $\mathcal{D} = \{(x_i, Y_i)\}_{i=1}^{|\mathcal{D}|}$: training dataset
- N_{tot} : the number of training rounds
- p, n : the number of positive and negative proposals
- $\bar{\rho}$: the IoU threshold

Output: the trained parameters θ_e^N, θ_d^N

```

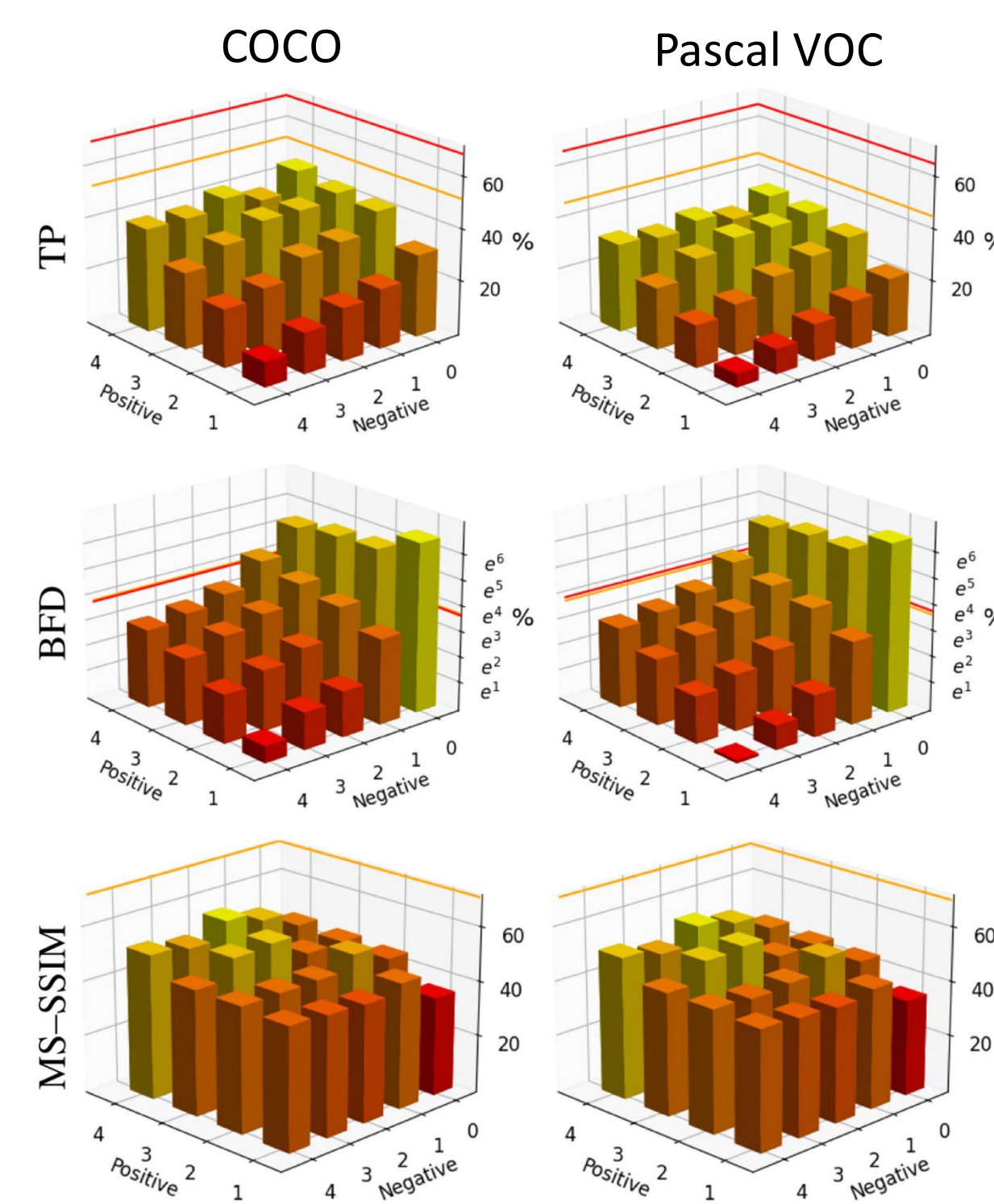
1:  $\theta_e^0, \theta_d^0 \leftarrow \text{RandInit}()$ 
2: for  $\tau \leftarrow 0$  to  $N_{tot}$  do
3:    $(x, Y) \sim \mathcal{D}$ 
4:    $\tilde{Y} = f(q(x; \theta_e^\tau, \theta_d^\tau); \theta_f)$ 
5:    $\tilde{Y}^{SEL} = \text{SELECTPROPOSALS}(Y, \tilde{Y}, p, n)$ 
6:    $\theta_e^{\tau+1}, \theta_d^{\tau+1} \leftarrow \text{BACKPROP}(\mathcal{L}_{TaskNet}(Y, \tilde{Y}^{SEL}), \theta_e^\tau, \theta_d^\tau)$ 
7: end for
8: procedure SELECTPROPOSALS( $Y_{GT}, Y_t, p, n$ )
9:    $S \leftarrow \emptyset$ 
10:  for all  $g \in Y_{GT}$  do
11:     $P \leftarrow \{t \in Y_t \mid \arg \max_{g \in Y_{GT}} \rho(t, g) = g\}$ 
12:     $P^{SEL} \leftarrow \{t_1, \dots, t_p \mid t_i \in P, \rho(t_i, g) \geq \rho(t_j, g), \forall j \geq i\}$ 
13:     $N \leftarrow \{t \in Y_t \setminus P \mid \arg \max_{g \in Y_{GT}} \rho(t, g) = g, \rho(t, g) < \bar{\rho}\}$ 
14:     $N^{SEL} \leftarrow \{t_1, \dots, t_n \mid t_i \in N, \sigma(t_i) \geq \sigma(t_j), \forall j \geq i\}$ 
15:     $S \leftarrow S \cup P^{SEL} \cup N^{SEL}$ 
16:  end for
17:  return  $S$ 
18: end procedure
  
```

Evaluation

Experimental setup

- People and vehicle detection (highly privacy-critical) using an off-the-shelf Faster R-CNN with a ResNet-50 backbone
- Trained and validated using filtered indoor contexts of the COCO dataset and out-of-distribution validation sets via Pascal VOC 2012
- Detection accuracy measured via AP , AP_{50} true positives (TP) and background false detections (BFD)
- Privacy measured via Multi-Scale Structural Similarity (MS-SSIM) of reconstructions from an optimized MIA attacker using Mean Absolute Error (MAE) and fine-grained Edge-Centric (EC) Sobel loss functions^[3]
- Baselines
 - training with all proposals (ALL), as done in [2]
 - TaskNet with plain data (TN)

Datasets validation



TaskNet (TN)
Baseline (ALL)

- ✓ Increasing Positive Proposals improves correct detections
- ✓ More Negative Proposals suppresses background detections
- ✓ Negative and positive proposals should be equal to have a good balance between correct detections and error suppression

Attacker ablation

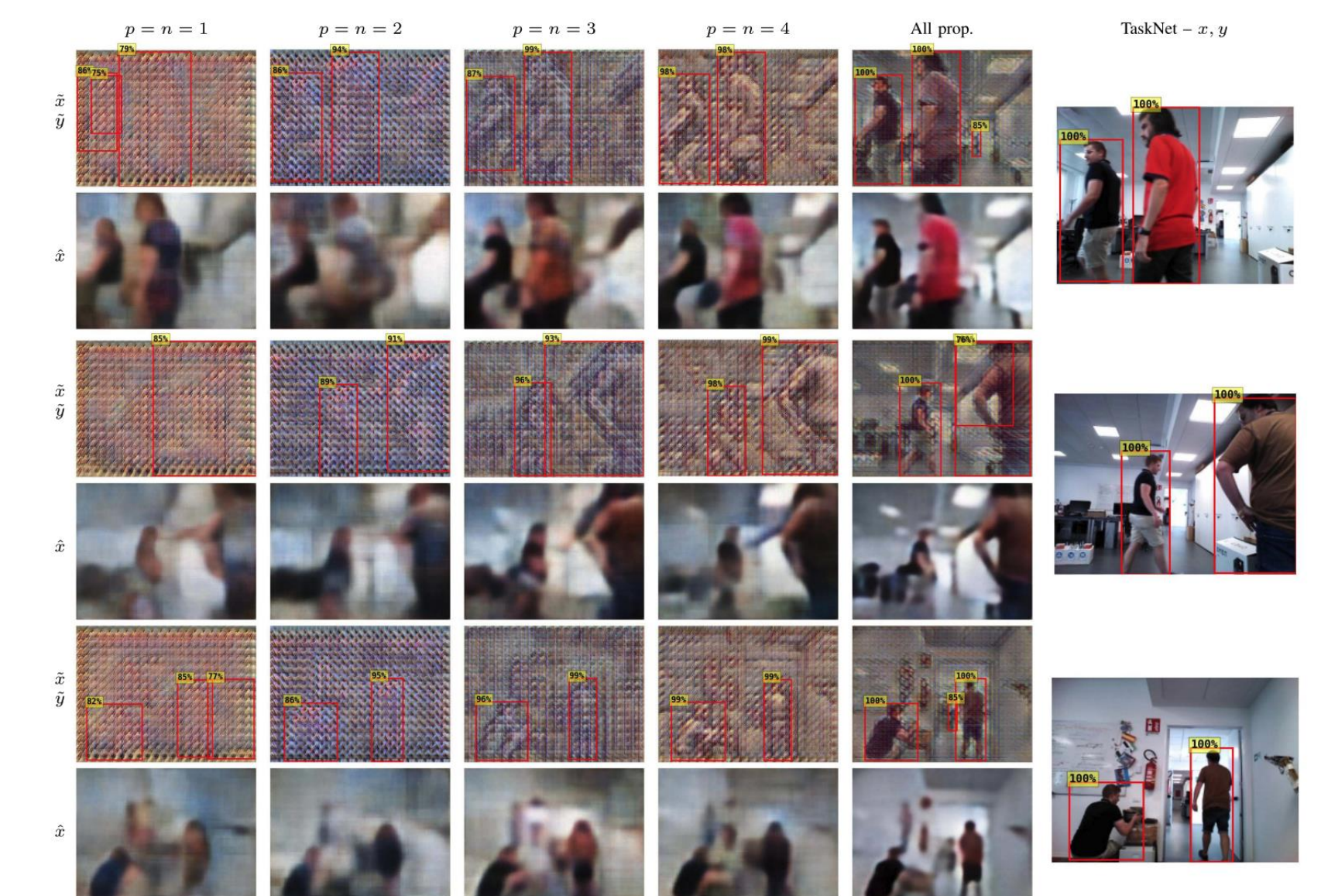


- ✓ A more powerful attacker (enhanced reconstruction loss) fails to restore privacy-sensitive features



p, n	AP $\uparrow$	AP ₅₀ $\uparrow$	TP $\uparrow$	BFD $\downarrow$	MS $\downarrow$
1, 1	37	68	58	8	53
2, 2	56	83	67	5	54
3, 3	57	84	77	10	61
4, 4	59	85	80	11	61
ALL	66	87	86	17	81
TN	78	96	95	19	100

Real-world experiment



References

- [1] Taras et al., "Inherently privacy-preserving vision for trustworthy autonomous systems: Needs and solutions," Journal of Responsible Technology, 2024
- [2] Nakanoya et al., "Co-design of communication and machine inference for cloud robotics," Autonomous Robots, 2023
- [3] Azizian et al., "Privacy-preserving autoencoder for collaborative object detection," IEEE Transactions on Image Processing, 2024

Introduction

Context

- We consider a fleet of robots deployed in different indoor environments that need to perform object detection
- This ability is essential to carry out high-level tasks useful in several contexts^[1]

Service Robots



Assistive Robots

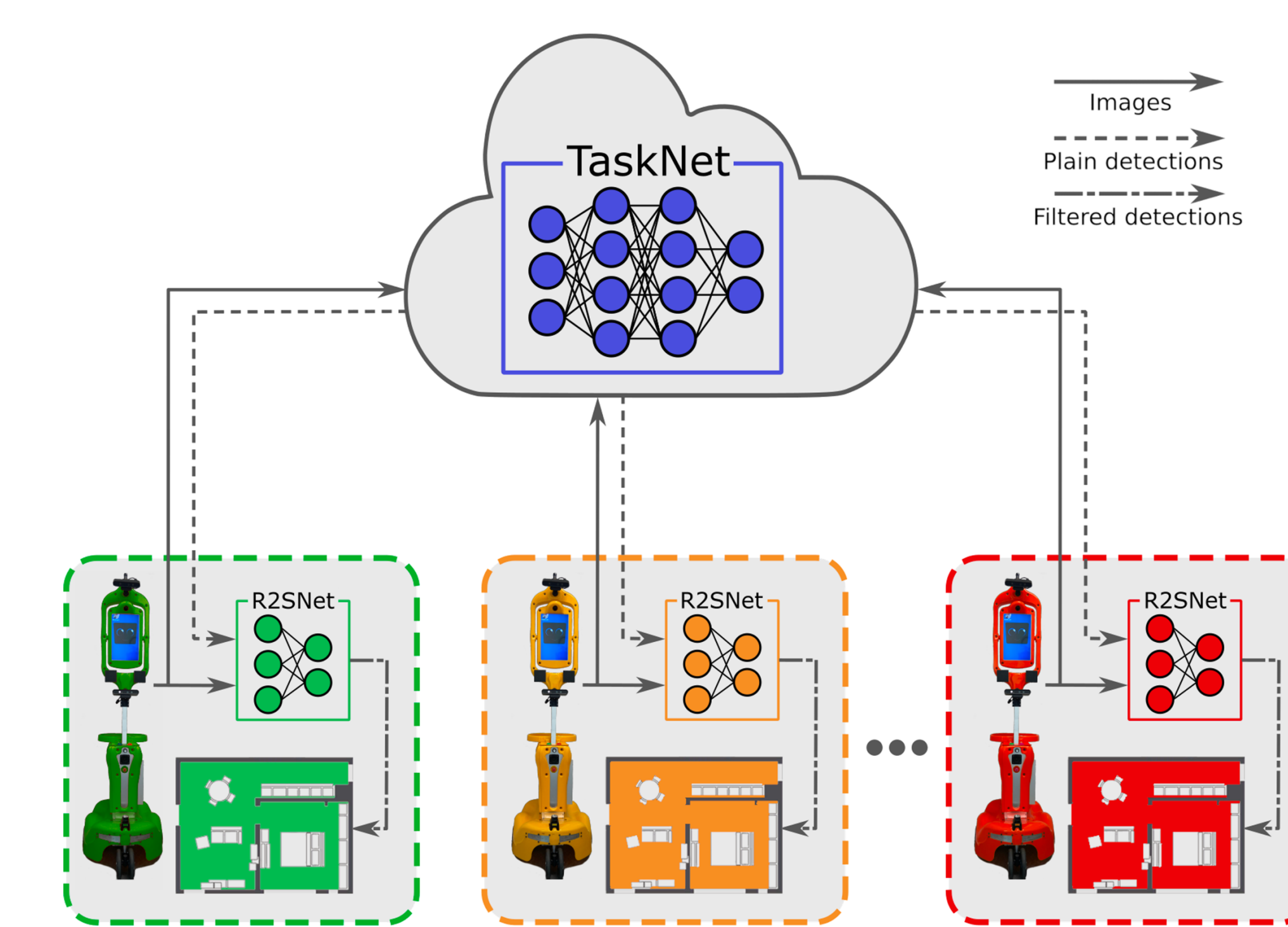


Robots as Computationally Limited Autonomous Agents

- A straightforward approach is to plug and play publicly-available Deep Neural Networks (DNNs) for object detection (OD)
- Running deep learning-based models on mobile robots is prohibitive
 - Low-powered and affordable hardware configuration
 - Limited computational capabilities affect real-time inference
 - Energy-preservation constraints for long-term autonomy

Cloud Robotics

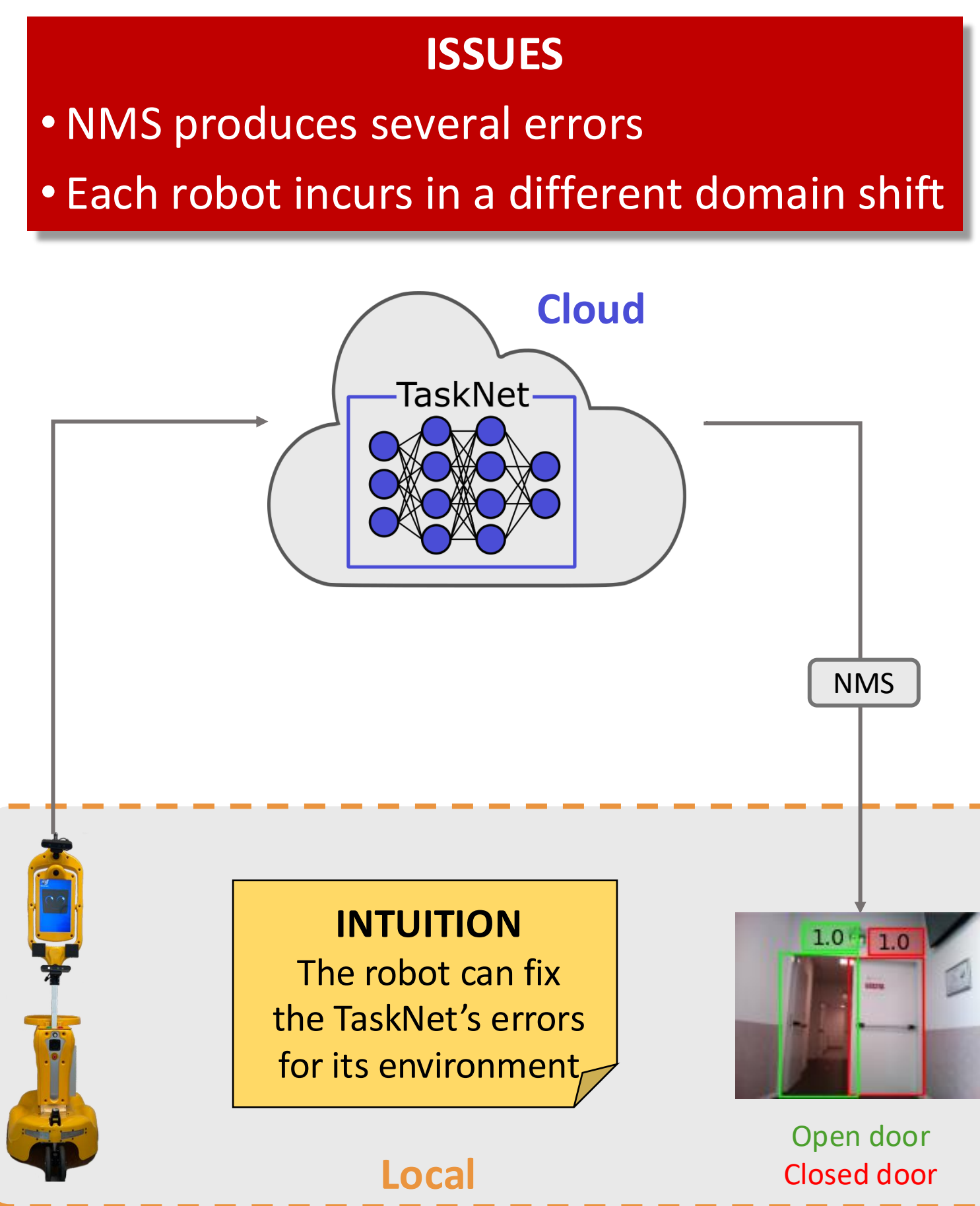
- Offloading computationally intensive inference tasks to third-party cloud services running DNNs, here called TaskNets^[2]
- Domain shift degrades the TaskNet's performance
- Classical domain adaptation^[3] cannot be applied
 - The TaskNet is inaccessible
 - Train, deploy, and maintain a TaskNet for each robot is expensive



Preliminaries

Object Detection over the Cloud

- The robot sends remotely its perceptions (RGB images)
- The TaskNet predicts a dense set of object proposals $\hat{Y} = \{\hat{y}\}$
- Bounding boxes are expressed as $\hat{y} = [\hat{c}_x, \hat{c}_y, \hat{w}, \hat{h}, \hat{c}, \text{hot}(\hat{o})]$
 - $\hat{c}_x, \hat{c}_y$ are the center coordinates
 - $\hat{w}, \hat{h}$ are width and height
 - $\hat{c}, \text{hot}(\hat{o})$ are the confidence and the one-hot encoded label
- $\hat{Y}$ is filtered using Non-Maximum Suppression (NMS)
- The remaining bounding boxes are sent back to the robot



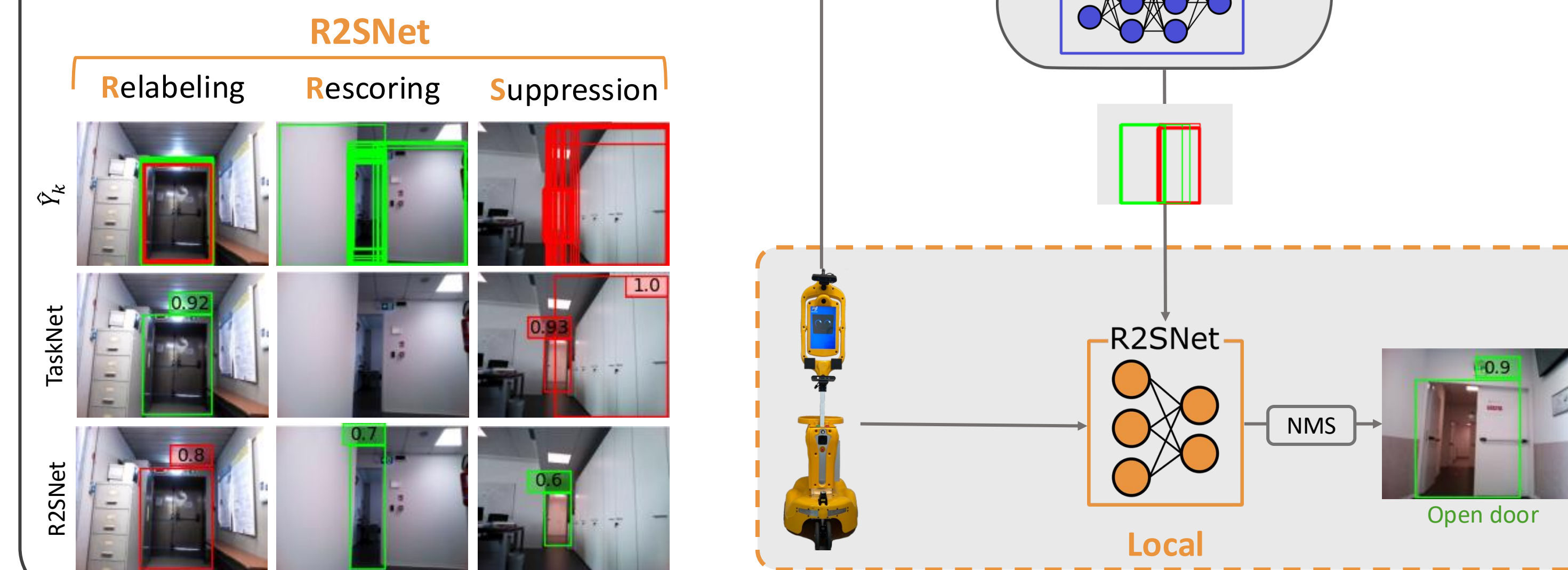
Approach

Downstream Proposal Refinement

- The robot receives $\hat{Y}$ and selects the first k most confident, obtaining $\hat{Y}_k$
- It refines their parameters with a lightweight DNN which performs 3 corrective actions
- $\hat{Y}_k$ is then filtered with NMS

BENEFITS

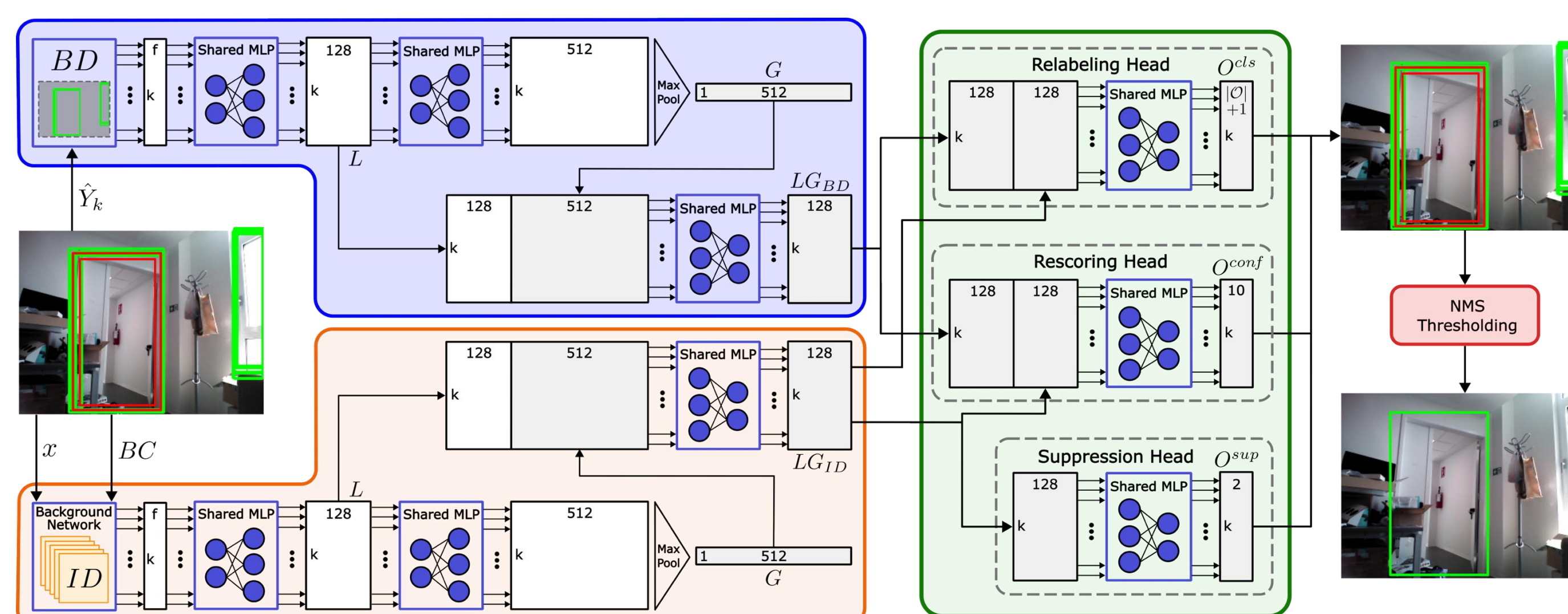
- Horizontally-scalable adaptation
- Computationally affordable by robots



Architecture

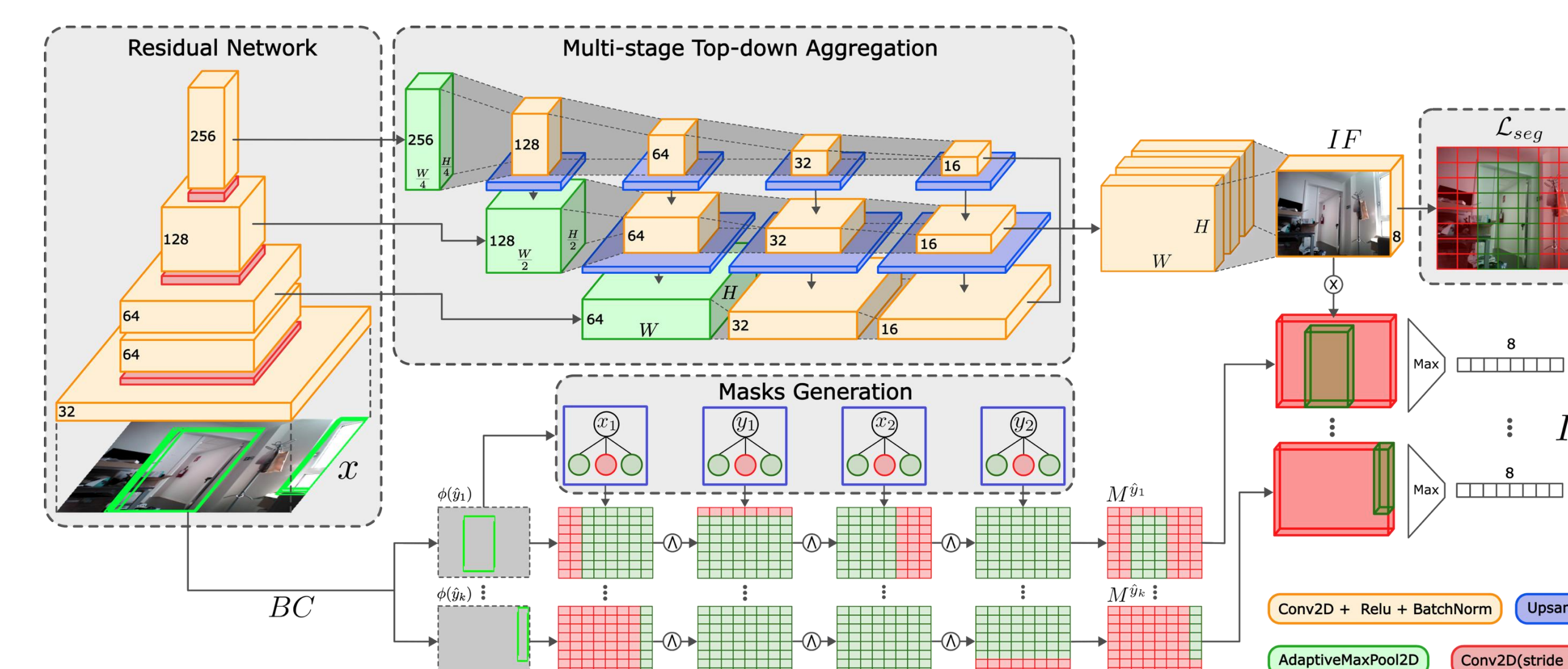
R2SNet Architecture

- Bounding boxes are expressed with two different descriptors:
 - Bounding-box Descriptors (BD): parameters of proposals received by the TaskNet
 - Image Descriptors (ID): visual features extracted by the Background Feature Network (BFNet)
- BD and ID are processed by two symmetric networks inspired by PointNet^[4]
 - Local features (L) are extracted through shared MLPs and Global features (G) with a $\max$ operator
 - Local and global features are then concatenated and mixed with shared MLPs in an embedding LG
- The mixed features are fed into 3 heads to perform relabeling, rescoring, and suppression



BFNet Architecture

- Produces an image feature map IF with dimension $[W, H, 8]$
 - Extracts a multi-scale embeddings using a residual network
 - The last 3 levels are processed by 3 parallel convolutional networks and top-down aggregated
- Produces a binary masks M for each proposal
 - 4 MLPs with fixed weights and biases
 - Each MLP extracts a partial mask for each coordinate that are aggregated with an and operator
- Masks are multiplied with IF and then maxpooled obtaining visual features for each proposals



Evaluation

- D_{DD2} : a real dataset (called DeepDoors2) with $\approx 3k$ examples
- D_G : photorealistic dataset obtained with

Train the TaskNet (Faster R-CNN)

Exp.	Mean			
	mAP $\uparrow$	TP $\uparrow$	FP $\downarrow$	BFD $\downarrow$
TaskNet	30	36%	7%	20%

Performance increases even with a few data

Ablation study			All heads contribute to domain adaptation			
Rel.	Res.	Sup.	mAP $\uparrow$	TP $\uparrow$	FP $\downarrow$	BFD $\downarrow$
✓			34	44%	10%	35%
	✓		44	48%	4%	6%
		✓	41	54%	15%	34%

